

EKLAVYA GOYAL

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Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science and Engineering

CGPA: 9.69 / 10.0 (4 semesters) • 3rd Year, 5th Semester

Vellore, India

Aug 2024 – May 2028

Class XII (CBSE): 95.8% • Class X (GSEB): 94.0%

Technical Skills

Languages	Java, C++, C, Python, JavaScript, SQL, HTML5, CSS3
Backend	Spring Boot, REST API design, Testing Services with Postman
Mobile	Android SDK (Java), Bluetooth Low Energy (BLE), OSMDroid maps, Firebase (Firestore)
Databases	Firebase Firestore (NoSQL); MySQL / SQL — DBMS coursework (schema design, normalization, indexing)
DevOps	Git, GitHub, Docker, Kubernetes, Jenkins, Maven, Nagios, CI/CD, Linux (CLI), IntelliJ IDEA, Android Studio, VS Code

Projects

MeshSOS — Offline Peer-to-Peer Emergency Network

Feb 2026 – Present

Java, Android SDK, BLE, OSMDroid, Firebase (Firestore)

github.com/EklavyaGoyal17/MeshSOS

Phones relay SOS alerts to each other over Bluetooth when cellular networks are down.

- Built an offline SOS network that relays distress alerts **phone-to-phone over Bluetooth Low Energy** with no internet or SIM — each device rebroadcasts alerts it receives, so a single message hops across nearby phones to reach help.
- Packed each alert into a compact **17-byte payload** (ByteBuffer: user ID, GPS, timestamp, status) carried in a connectionless BLE manufacturer-data advertisement, removing any need for device pairing.
- Prevented rebroadcast storms with a **deduplication cache** (keyed on sender ID + timestamp) and a **15-second relay cutoff**, so each alert propagates once instead of looping through the mesh.
- Added a gateway-to-cloud bridge that mirrors received alerts to a **Firebase** dispatch dashboard, plus an offline **OSMDroid** victim map, a **112** no-SIM failsafe dialer, and runtime localization across **7 Indian languages**.

Smart City IAQ Node — Backend & System Architecture (Team Lead)

2026

Node.js, Express, SerialPort, Axios, REST, Open-Meteo API

github.com/EklavyaGoyal17/SmartCity-IAQ-Node

Estimates a room's air quality from crowd density and auto-triggers ventilation — no gas sensors.

- Led a 2-member team and designed the backend for an IoT node that estimates indoor air quality from **room occupancy** (ultrasonic sensor count) instead of costly chemical gas sensors.
- Built a **Node.js** service that ingests the Arduino's serial JSON stream, maintains a rolling **20-point history** buffer, and exposes it to a React dashboard via a **REST** endpoint.
- Implemented closed-loop actuation logic that triggers a ventilation relay when **AQI $\geq$ 300** (EPA hazardous threshold), driven directly by live sensor input.
- Enriched local telemetry with ambient temperature and humidity from the **Open-Meteo REST API**, polled on a fixed 10-minute interval to limit outbound requests.

techportal — Containerized CI/CD Deployment Pipeline

2026

Spring Boot, Docker, Kubernetes, Jenkins, Maven, Nagios

github.com/EklavyaGoyal17/techportal

Automates a Spring Boot service from Git commit to a live, monitored Kubernetes deployment.

- Built a **Spring Boot** REST service (Java, Maven) and containerized it with **Docker** on an Eclipse Temurin JDK base image.
- Authored a **4-stage declarative Jenkins pipeline** (checkout → Maven build → Docker build → Kubernetes deploy) that tags each image with the build ID and injects it into the manifest for versioned, repeatable rollouts.
- Wrote **Kubernetes** manifests — a **2-replica** Deployment and a NodePort Service — and configured **Nagios** HTTP monitoring against the exposed node port.
- Instrumented the service with **Spring Actuator** and **Micrometer**, exporting runtime metrics to **Graphite** for live health and performance observability.

Achievements & Certifications

• **Selected — JPMorganChase Code for Good 2026**

Aug 2026

Chosen for JPMorganChase's flagship 24-hour software engineering hackathon (Bengaluru), building technology solutions for nonprofits alongside JPMorganChase engineers.

• **DevOps, Agile and Design Thinking** — IBM Career Education Program.

Jul 2026

• **Problem Solving: 150+ DSA problems** solved on LeetCode (arrays, strings, trees, graphs, dynamic programming, sliding window); **5-star HackerRank** badges in **Java, Python, and SQL**.

Coursework

Data Structures & Algorithms • Object-Oriented Programming • Database Management Systems • Operating Systems

• Computer Networks • Computer Architecture • Theory of Computation • Software Engineering (Agile / DevOps)